



Department of Electrical Electronics Engineering

Academic Year 2021 – 2022(Odd Semester)

Degree, Semester & Branch: B.E/III /EEE-B

Course Code & Title: EC8353/Electron Devices and Circuits

Name of the Faculty member (s): Mrs.G.Sivapriya

Innovative Practice Description

- **Unit / Topic:** Unit I / Applications of Diode
- **Course Outcome:** CO 1- Explain the structure and operation of basic electronic devices such as diodes
- **Topic Learning Outcome:** TLO1.
- **Activity Chosen:** One Minute Paper
- **Justification:**

The structure, operation and V-I characteristics of PN junction diode. After teaching this topic ,I thought of this activity for making to have clarity about the operation of the diode and its characteristics. I can judge the understanding level of the students .

- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**
 - After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.
 - Total Strength :30
 - Reporter: Myself
 - At the end of the Class (Last 5 minutes)
 - I asked the students to think about the diode operation and its applications concept for 3 minutes.
 - Then I told them to write as much as they can within a short period of time (1 minute)
 - Finally, I collected the papers from each column.

- **CO – PO / PSO mapping:**

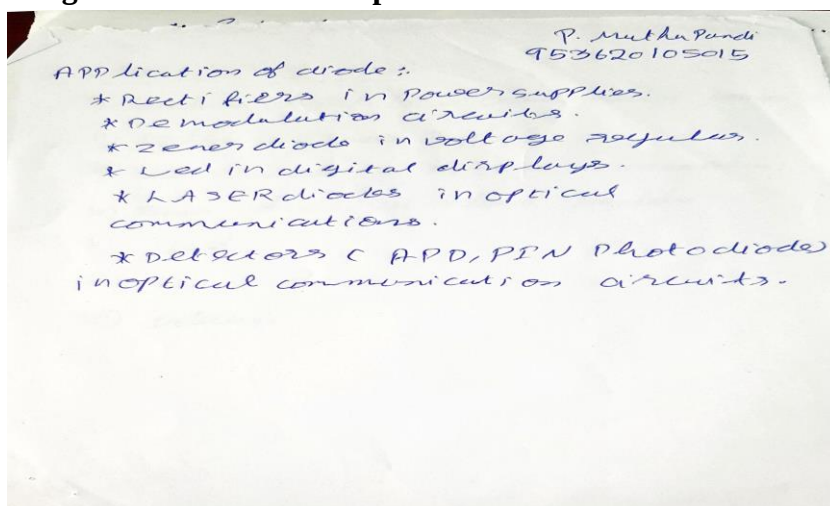
CO	PO1	PO2	PO3	PO4	PO5	PO8	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	1	1	1	1	1	2	2	2

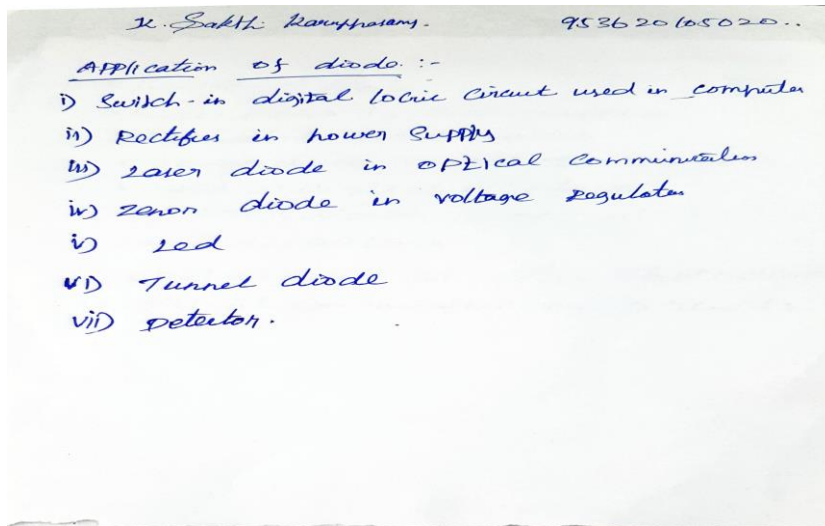
(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO8	PO10	PO12	PSO1	PSO2	PSO3
	3	3	2	1	1	1	1	1	2	2	2
Justification for correlation	Students will be solving basic problem on Semiconductor devices using basic electrical concepts.	Suitable mathematical and engineering knowledge will be applied by the students to solve basic problems	The electronics application needs analysis of characteristics of semiconductor devices to solve problems related to improve the gain.	Students will do the experiments to obtain the characteristic of semiconductor devices.	Students will be using Electronics virtual laboratory and gains proficiency on the same	Students follow basic ethical practice to prepare the assignment.	Assignments given will ensure the submission of a technical report with structured flow.	For preparing the assignment, students need to comprehend few technical contents.	The basic concepts on semiconductor devices are essential to analyze the performance of electronic circuits	For preparing the assignment, students need to comprehend few technical contents	Simulation/Design knowledge on semiconductor devices operation is essential to design the hardware requirement in industrial automation

• Images / Screenshot of the practice:





- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- The assessment of effectiveness of the activity was felt when told most of the points.
- While conducting the activity, I understood that the students will be able to explain the applications of diodes.
- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 90% of students answered.

- ❖ **Benefit of the practice:**

- Students were able to attend the question even in the questions are in indirect form.
- Students were able to explain the concepts in examination without any confusion.

- ❖ **Challenges faced in implementation:**

- In the class mostly girls hesitate to answer the questions.
- Time utilization for conducting activity.

References:

1. David A. Bell, "Electronic devices and circuits", Oxford University higher education, 5th edition 2008.
2. Sedra and Smith, "Microelectronic circuits", 7th Edition, Oxford University Press.

Signature of Faculty Member

Form No. AC 10c

HOD

Rev.No.01

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Innovative Practice Description

- **Unit / Topic:** Unit II / Characteristics of BJT under CE configuration
- **Course Outcome:** CO 2- Illustrate the characteristics of different electronic devices such as transistors and thyristors.
- **Topic Learning Outcome:** TLO4.
- **Activity Chosen:** Virtual lab demonstration.
- **Justification:**

The structure, operation and V-I characteristics of transistor. After teaching this topic ,I thought of this activity for making to have clarity about the operation of the transistor and its characteristics very easily. I can judge the understanding level of the students .

- **Time Allotted for the Activity:** 5 Minutes

- **Details of the Implementation:**

- After teaching the concept, I opened virtual lab software and explained how to give connections, how to obtain the characteristics of transistor in CE configuration. Finally, I taught to students how to draw the input and output characteristics of CE configuration.
- Total Strength :30
- Reporter: Myself
- At the end of the Class (Last 5 minutes)
- I asked the students to obtain the characteristics of transistor in CE configuration in virtual lab.
- Finally, I told the students to send the report of that experiment.

- **CO – PO / PSO mapping:**

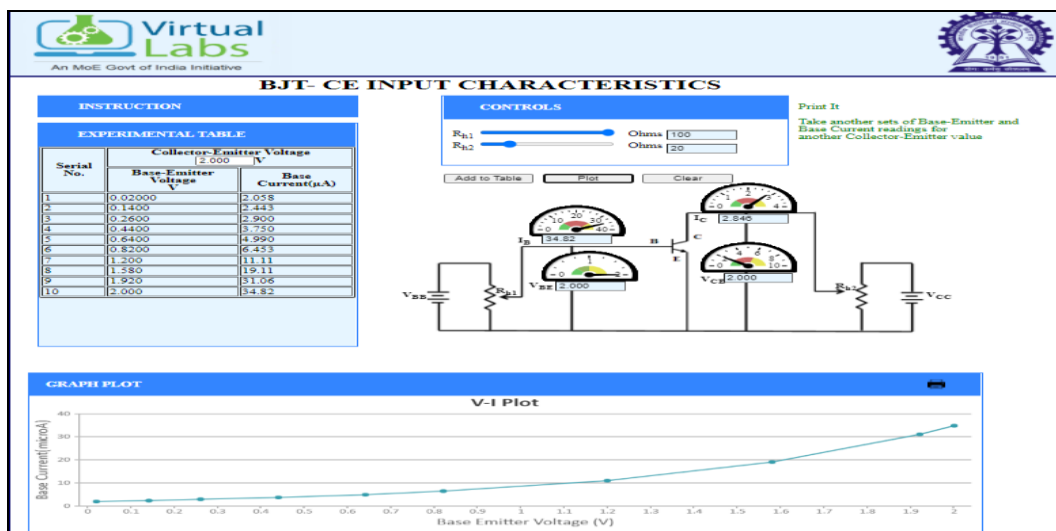
CO	PO1	PO2	PO3	PO4	PO5	PO8	PO10	PO12	PSO2
CO2	3	3	2	1	1	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

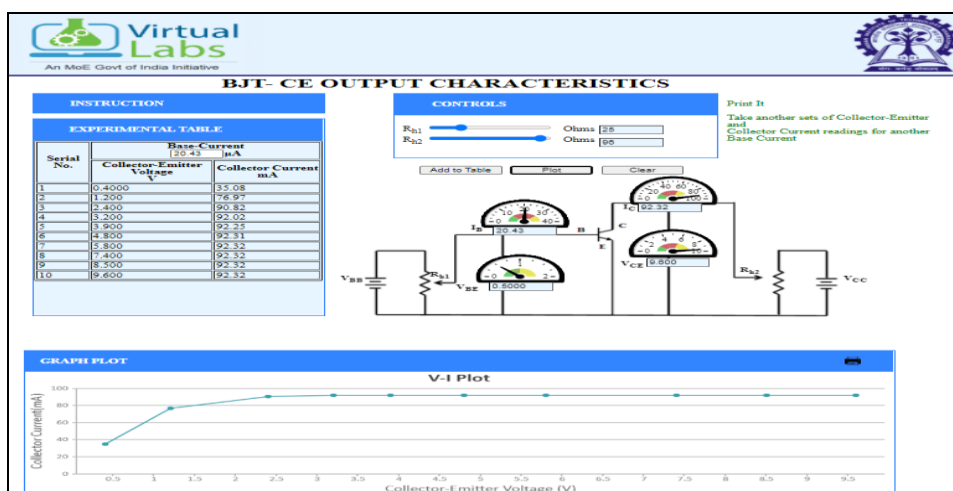
- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO8	PO10	PO12	PSO2
	3	3	2	1	1	1	1	1	2
Justification for correlation	Students will be solving basic problem on Semiconductor devices using basic electrical concepts.	Suitable mathematical and engineering knowledge will be applied by the students to solve basic problems	The electronics applications needs analysis of characteristics of semiconductor devices to solve problems related to improve the gain .	Students will do the experiments to obtain the characteristic of semiconductor devices.	Students will be using Electronics virtual laboratory and gains proficiency on the same	Students follow basic ethical practice to prepare the assignment .	Assignments given will ensure the submission of a technical report with structured flow.	For preparing the assignment, students need to comprehend few technical contents.	For preparing the assignment, students need to comprehend few technical contents

- Images / Screenshot of the practice:
Input Characteristics of BJT under CE Configuration



Output Characteristics of BJT under CE Configuration



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- The assessment of effectiveness of the activity was felt when told most of the points.
- While conducting the activity, I understood that the students will be able to understand the CE configuration of transistor.
- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 85% of students answered.

Benefit of the practice:

- Students were able to attend the question even in the questions are in indirect form.
- Students were able to explain the concepts in examination without any confusion.

❖ *Challenges faced in implementation:*

- In the class some students hesitate to do this in virtual lab.
- Time utilization for conducting activity.

References:

1. David A. Bell ,”Electronic devices and circuits”, Oxford University higher education, 5th edition 2008.
2. Sedra and smith, “Microelectronic circuits”, 7th Edition, Oxford University Press.

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Innovative Practice Description

- **Unit / Topic:** Unit V / Oscillators
- **Course Outcome:** CO5- Perform design and analysis of feedback amplifiers and oscillators.
- **Topic Learning Outcome:** TL14,TLO15
- **Activity Chosen:** Ungraded Quiz
- **Justification:**

The expression for frequency of oscillation of various oscillators. After teaching this topic, I thought of this activity for making to have clarity about the operation of the oscillator and its characteristics very easily. I can judge the understanding level of the students .

- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**
 - After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.
 - Total Strength :29
 - Reporter: Myself
 - At the end of the Class (Last 10 minutes)
 - I asked the students to think about the Oscillator working principle, types of oscillators and the frequency of oscillator for 2 minutes.
 - Then I told them to write to attend the quiz which is ungraded(8 minutes)
 - Finally, I collected the responses from each students.
- **CO – PO / PSO mapping:**

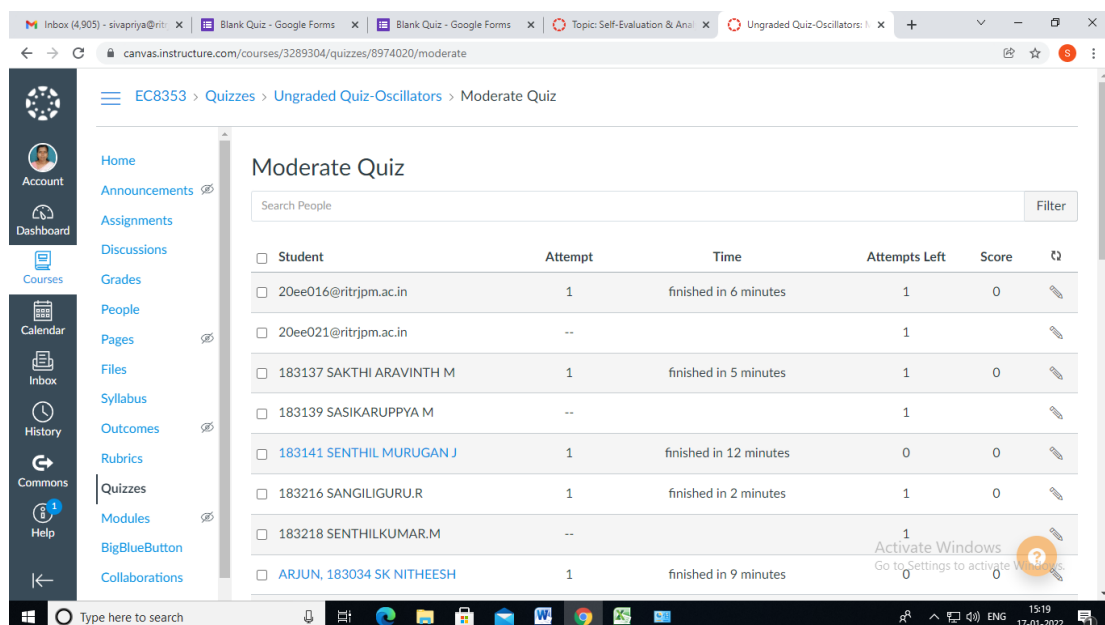
CO	PO1	PO2	PO3	PO4	PO5	PO8	PO10	PO12	PSO2
CO2	3	3	2	1	1	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO8	PO10	PO12	PSO2
	3	3	2	1	1	1	1	1	2
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- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

I preferred this activity because it will help me to evaluate what types of questions I am asking students and what level they are working at.

Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students can understand their knowledge in that particular topic .

- ❖ ***Challenges faced in implementation:***

- Since it is conducted online ,it is very tough to make all students to attend.
 - It is difficult to access whether student have attended the quiz properly or not.

References:

- ❖ https://www.nactateachers.org/images/The_Student-Developed_Quiz_or_Exam-__Scaffolding_Higher-Order_Thinking.pdf
- ❖ David A. Bell ,”Electronic devices and circuits”, Oxford University higher education, 5th edition 2008.
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Innovative Practice Description

- **Unit / Topic:** Unit IV / Neutralization methods
- **Course Outcome:** CO4- Explore the working of multistage, differential and power amplifiers
- **Topic Learning Outcome:** TL11
- **Activity Chosen:** Class Poll
- **Justification:**

The need and types of neutralization methods. After teaching this topic, I thought of this activity for making to have clarity about the need of Neutralization and types of neutralization methods very easily. I can judge the understanding level of the students .

- **Time Allotted for the Activity:** 5 Minutes
- **Details of the Implementation:**
 - After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.
 - Total Strength :30
 - Reporter: Myself
 - At the end of the Class (Last 10 minutes)
 - I asked the students to think about the need and types of neutralization methods for 2 minutes.
 - Then I told them to attend the two Google quiz which is graded(8 minutes)
 - Finally, I collected the responses from each students.
- **CO – PO / PSO mapping:**

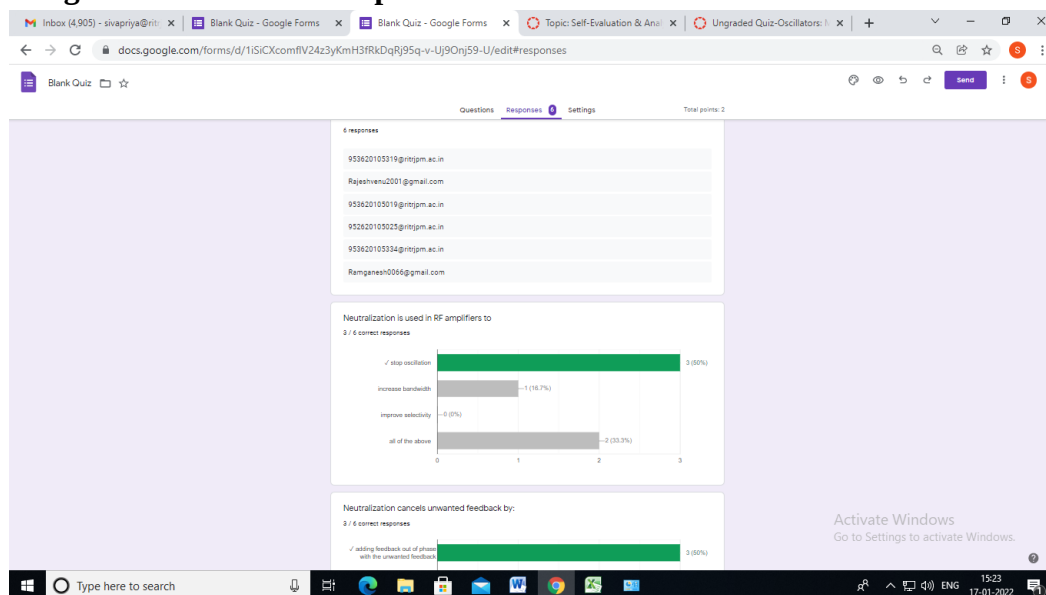
CO	PO1	PO2	PO3	PO4	PO5	PO8	PO10	PO12	PSO2
CO2	3	2	2	1	1	1	1	1	2

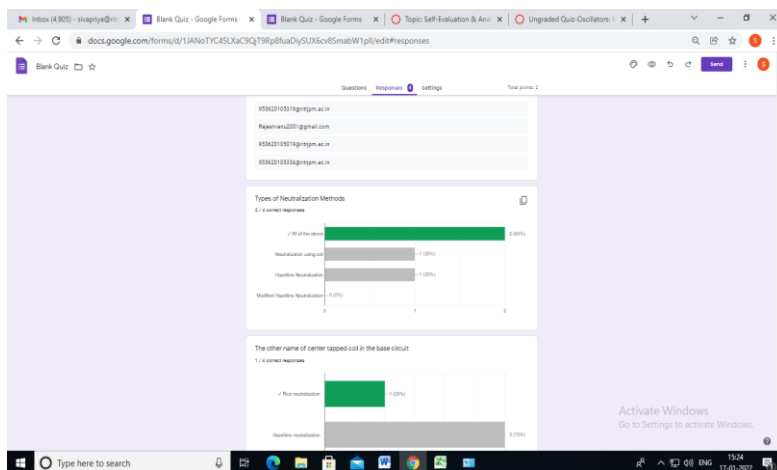
(1 – Low 2 – Moderate 3 – High)

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- Images / Screenshot of the practice:





• Reflective Critique:

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